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5 A ray of light is incident on a soap film. Fig. 5.1 shows a magnified image of a small part of the soap film. The ray of light is refracted as it enters the soap film.

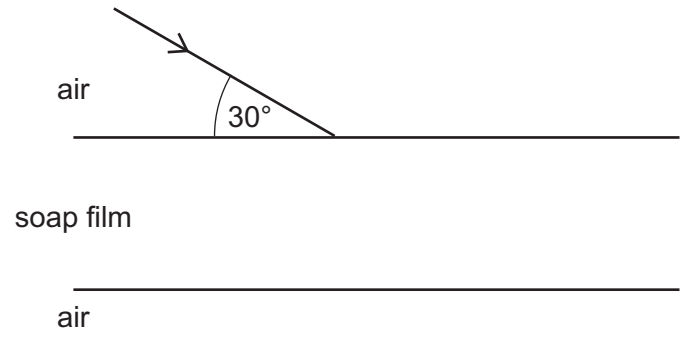

Fig. 5.1

The refractive index of the soap film is 1.28.

(a) Define refractive index in terms of the speed of light.

.....

..... [1]

(b) **(i)** Show that the angle of refraction as the light enters the soap film is approximately 43°.

[2]

(ii) On Fig. 5.1, carefully draw the refracted light ray in the soap film and label the angle of refraction. [2]

(c) The ray of light is monochromatic red light with a wavelength of 680 nm in air.

(i) Define monochromatic.

.....

..... [1]

(ii) Calculate the frequency of the light.

frequency = ... [3]

[Total: 9]